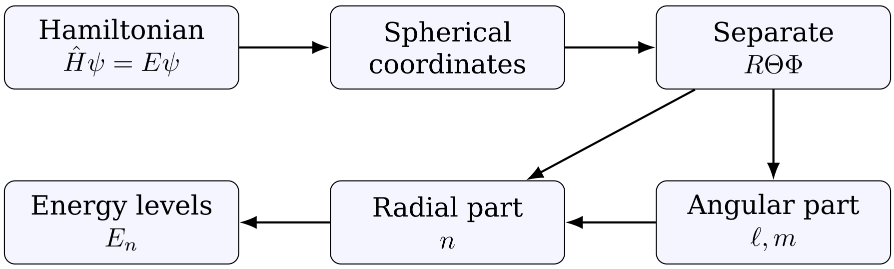
Solving the Schrödinger Equation for the Hydrogen Atom

Separation of Variables, Quantum Numbers, and Energy Quantization

Technical lecture notes for computer science students



Purpose of these notes

These notes show the actual technical path from the stationary Schrödinger equation to the hydrogen atom eigenstates:

$$\psi_{n\ell m}(r,\theta,\varphi) = R_{n\ell}(r)Y_\ell^m(\theta,\varphi), \qquad E_n = -\frac{13.6\,\mathrm{eV}}{n^2}.$$

The goal is not to prove every theorem about special functions. The goal is to make every separation step visible: where the constants come from, why quantum numbers appear, and why the energy becomes discrete.

Contents

1 Problem setup

1.1 The physical system

The hydrogen atom consists of a proton and an electron. In the simplest approximation, the proton is placed at the origin and the electron moves in the Coulomb potential

$$V(r) = -\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r} = -\frac{\gamma}{r}, \qquad \gamma = \frac{e^2}{4\pi\varepsilon_0}.$$

For better accuracy, one replaces the electron mass by the reduced mass

$$\mu = \frac{m_e m_p}{m_e + m_p}.$$

For an introductory derivation, one may take $\mu \approx m_e$.

1.2 The stationary Schrödinger equation

For time-independent states we solve the eigenvalue problem

$$\hat{H}\psi = E\psi,$$

where the Hamiltonian is

$$\hat{H} = -\frac{\hbar^2}{2\mu}\nabla^2 - \frac{\gamma}{r}.$$

Therefore,

$$\left[-\frac{\hbar^2}{2\mu}\nabla^2 - \frac{\gamma}{r}\right]\psi = E\psi. \tag{1}$$

Computational analogy

This is an eigenvalue problem. In linear algebra we solve

$$Av = \lambda v.$$

Here we solve

$$\hat{H}\psi = E\psi.$$

Allowed energies E are eigenvalues. Wave functions ψ are eigenvectors, but in an infinite-dimensional function space.

2 Why spherical coordinates are used

The potential depends only on the distance from the nucleus:

$$V(r) = -\frac{\gamma}{r}.$$

It does not depend on direction. This spherical symmetry suggests using spherical coordinates

$$(r, \theta, \varphi),$$

where

$$r = \sqrt{x^2 + y^2 + z^2}.$$

The Laplacian in spherical coordinates is

$$\nabla^2\psi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\varphi^2}. \tag{2}$$

What students usually miss

The difficult part is not the Coulomb potential. The difficult part is the Laplacian. In Cartesian coordinates the potential is ugly. In spherical coordinates the Laplacian is longer, but the symmetry becomes visible.

3 Insert the spherical Laplacian

Start from Eq. (1):

$$-\frac{\hbar^2}{2\mu}\nabla^2\psi - \frac{\gamma}{r}\psi = E\psi.$$

Move the potential term to the right form:

$$-\frac{\hbar^2}{2\mu}\nabla^2\psi = \left(E + \frac{\gamma}{r}\right)\psi.$$

Multiply by $-2\mu/\hbar^2$:

$$\nabla^2\psi + \frac{2\mu}{\hbar^2}\left(E + \frac{\gamma}{r}\right)\psi = 0. \tag{3}$$

Now substitute Eq. (2) into Eq. (3):

$$\begin{aligned}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\psi}{\partial\varphi^2}& \\ + \frac{2\mu}{\hbar^2}\left(E + \frac{\gamma}{r}\right)\psi = 0.&\end{aligned} \tag{4}$$

This is the equation we now separate.

4 Separation ansatz

We look for product solutions:

$$\psi(r,\theta,\varphi) = R(r)\Theta(\theta)\Phi(\varphi). \tag{5}$$

This is not the most general solution immediately. It is a standard technique: first solve the separated eigenfunctions, then build general states as combinations.

Substitute $\psi = R\Theta\Phi$ into Eq. (4). Since each factor depends only on one variable,

$$\frac{\partial\psi}{\partial r} = \Theta\Phi\frac{\mathrm{d}R}{\mathrm{d}r}, \qquad \frac{\partial\psi}{\partial\theta} = R\Phi\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}, \qquad \frac{\partial^2\psi}{\partial\varphi^2} = R\Theta\frac{\mathrm{d}^2\Phi}{\mathrm{d}\varphi^2}.$$

After substitution:

$$\frac{\Theta\Phi}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}R}{\mathrm{d}r}\right)+\frac{R\Phi}{r^2\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right)+\frac{R\Theta}{r^2\sin^2\theta}\frac{\mathrm{d}^2\Phi}{\mathrm{d}\varphi^2}$$
$$+\frac{2\mu}{\hbar^2}\left(E+\frac{\gamma}{r}\right)R\Theta\Phi=0.$$

Divide by $R\Theta\Phi$:

$$\frac{1}{Rr^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}R}{\mathrm{d}r}\right)+\frac{1}{\Theta r^2\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right)+\frac{1}{\Phi r^2\sin^2\theta}\frac{\mathrm{d}^2\Phi}{\mathrm{d}\varphi^2}$$
$$+\frac{2\mu}{\hbar^2}\left(E+\frac{\gamma}{r}\right)=0.$$

Multiply by r^2:

$$\frac{1}{R}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}R}{\mathrm{d}r}\right)+\frac{2\mu}{\hbar^2}\left(Er^2+\gamma r\right)+\frac{1}{\Theta\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right)+\frac{1}{\Phi\sin^2\theta}\frac{\mathrm{d}^2\Phi}{\mathrm{d}\varphi^2}=0. \tag{6}$$

5 First separation: radial part versus angular part

Equation (6) has the structure

$$\underbrace{F(r)}_{\text{depends only on } r}\quad+\quad\underbrace{G(\theta,\varphi)}_{\text{depends only on angles}}\quad=0.$$

The only way this can hold for all r,θ,φ is that both parts are constants. We write the separation constant as $\ell(\ell+1)$. This notation is chosen because it gives the standard angular momentum spectrum.

Set

$$\frac{1}{R}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}R}{\mathrm{d}r}\right)+\frac{2\mu}{\hbar^2}\left(Er^2+\gamma r\right)=\ell(\ell+1), \tag{7}$$

and therefore

$$\frac{1}{\Theta\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right)+\frac{1}{\Phi\sin^2\theta}\frac{\mathrm{d}^2\Phi}{\mathrm{d}\varphi^2}=-\ell(\ell+1). \tag{8}$$

Rule of separation constants

When one side depends only on one variable and the other side depends only on another variable, both sides must be constant. Those constants become quantum numbers after boundary conditions are imposed.

6 Second separation: φ and θ

Take Eq. (8) and multiply by $\sin^2\theta$:

$$\frac{\sin\theta}{\Theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta\frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right)+\ell(\ell+1)\sin^2\theta+\frac{1}{\Phi}\frac{\mathrm{d}^2\Phi}{\mathrm{d}\varphi^2}=0. \tag{9}$$

The last term depends only on φ, while the rest depends only on θ. Therefore,

$$\frac{1}{\Phi}\frac{\mathrm{d}^2\Phi}{\mathrm{d}\varphi^2}=-m^2. \tag{10}$$

This gives the φ-equation:

$$\frac{\mathrm{d}^2\Phi}{\mathrm{d}\varphi^2} + m^2\Phi = 0. \tag{11}$$

Its solutions are

$$\Phi(\varphi) = A\mathrm{e}^{\mathrm{i}m\varphi} + B\mathrm{e}^{-\mathrm{i}m\varphi}.$$

Usually we take one complex basis function:

$$\Phi_m(\varphi) = \mathrm{e}^{\mathrm{i}m\varphi}.$$

6.1 Boundary condition in φ

The physical wave function must be single-valued after a full rotation:

$$\Phi(\varphi + 2\pi) = \Phi(\varphi).$$

For $\Phi = \mathrm{e}^{\mathrm{i}m\varphi}$, this gives

$$\mathrm{e}^{\mathrm{i}m(\varphi+2\pi)} = \mathrm{e}^{\mathrm{i}m\varphi} \quad \Longrightarrow \quad \mathrm{e}^{\mathrm{i}2\pi m} = 1.$$

Therefore,

$$\boxed{m = 0, \pm 1, \pm 2, \ldots} \tag{12}$$

This is the first quantum number we meet explicitly.

6.2 The θ-equation

Insert Eq. (10) into Eq. (8):

$$\frac{1}{\Theta \sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta \frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right) - \frac{m^2}{\sin^2\theta} = -\ell(\ell+1).$$

Multiplying by Θ, we obtain

$$\frac{1}{\sin\theta}\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\sin\theta \frac{\mathrm{d}\Theta}{\mathrm{d}\theta}\right) + \left[\ell(\ell+1) - \frac{m^2}{\sin^2\theta}\right]\Theta = 0. \tag{13}$$

This is the associated Legendre equation in angular form.

Its physically acceptable solutions are finite at the poles $\theta = 0$ and $\theta = \pi$. This condition allows only

$$\boxed{\ell = 0, 1, 2, \ldots, \qquad m = -\ell, -\ell+1, \ldots, \ell.} \tag{14}$$

Angular result

The angular part gives spherical harmonics:

$$Y_\ell^m(\theta, \varphi) = \Theta_\ell^m(\theta)\Phi_m(\varphi).$$

The quantum number ℓ describes the angular momentum magnitude, while m describes its projection on the chosen axis.

7 The radial equation

Return to Eq. (7):

$$\frac{1}{R}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}R}{\mathrm{d}r}\right)+\frac{2\mu}{\hbar^2}\left(Er^2+\gamma r\right)=\ell(\ell+1).$$

Multiply by R:

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}R}{\mathrm{d}r}\right)+\left[\frac{2\mu}{\hbar^2}\left(Er^2+\gamma r\right)-\ell(\ell+1)\right]R=0. \tag{15}$$

Divide by r^2:

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}R}{\mathrm{d}r}\right)+\left[\frac{2\mu}{\hbar^2}\left(E+\frac{\gamma}{r}\right)-\frac{\ell(\ell+1)}{r^2}\right]R=0. \tag{16}$$

This is the radial Schrödinger equation.

7.1 Removing the first-derivative structure

It is convenient to define

$$u(r)=rR(r),\qquad R(r)=\frac{u(r)}{r}. \tag{17}$$

A useful identity is

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}R}{\mathrm{d}r}\right)=\frac{1}{r}\frac{\mathrm{d}^2u}{\mathrm{d}r^2}.$$

Using $R=u/r$ in Eq. (16) and multiplying by r, we get

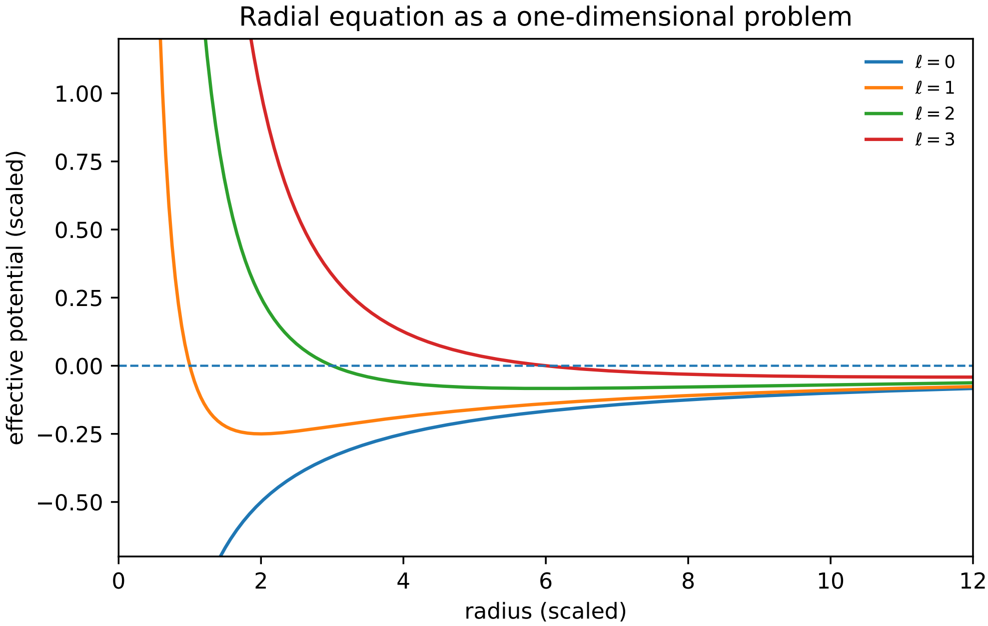
$$\frac{\mathrm{d}^2u}{\mathrm{d}r^2}+\left[\frac{2\mu}{\hbar^2}\left(E+\frac{\gamma}{r}\right)-\frac{\ell(\ell+1)}{r^2}\right]u=0. \tag{18}$$

One-dimensional interpretation

Equation (18) looks like a one-dimensional Schrödinger equation with an effective potential

$$V_{\text{eff}}(r)=-\frac{\gamma}{r}+\frac{\hbar^2\ell(\ell+1)}{2\mu r^2}.$$

The second term is the centrifugal barrier. It is zero for $\ell=0$ and pushes higher-ℓ states away from the origin.



8 Bound states and dimensionless variables

For hydrogen bound states, the energy is negative:

$$E < 0.$$

Define

$$\kappa = \frac{\sqrt{-2\mu E}}{\hbar}. \tag{19}$$

Then

$$\frac{2\mu E}{\hbar^2} = -\kappa^2.$$

Equation (18) becomes

$$\frac{\mathrm{d}^2 u}{\mathrm{d}r^2} + \left[-\kappa^2 + \frac{2\mu\gamma}{\hbar^2 r} - \frac{\ell(\ell+1)}{r^2}\right] u = 0. \tag{20}$$

Now define a dimensionless radius

$$\rho = 2\kappa r. \tag{21}$$

Then

$$\frac{\mathrm{d}}{\mathrm{d}r} = 2\kappa \frac{\mathrm{d}}{\mathrm{d}\rho}, \qquad \frac{\mathrm{d}^2}{\mathrm{d}r^2} = 4\kappa^2 \frac{\mathrm{d}^2}{\mathrm{d}\rho^2}.$$

Substituting into Eq. (20) and dividing by $4\kappa^2$ gives

$$\frac{\mathrm{d}^2 u}{\mathrm{d}\rho^2} + \left[-\frac{1}{4} + \frac{\eta}{\rho} - \frac{\ell(\ell+1)}{\rho^2}\right] u = 0, \tag{22}$$

where

$$\eta = \frac{\mu\gamma}{\hbar^2 \kappa}. \tag{23}$$

9 Asymptotic analysis: what the solution must look like

Before solving Eq. (22), analyze two limits.

9.1 Large radius: $\rho \to \infty$

For large ρ, the terms η/ρ and $\ell(\ell+1)/\rho^2$ become small. The equation becomes approximately

$$\frac{\mathrm{d}^2 u}{\mathrm{d}\rho^2} - \frac{1}{4}u \approx 0.$$

The solutions are

$$u \sim \mathrm{e}^{\rho/2} \quad \text{or} \quad u \sim \mathrm{e}^{-\rho/2}.$$

The growing solution $\mathrm{e}^{\rho/2}$ is not normalizable. Therefore physical bound states must contain

$$u(\rho) \sim \mathrm{e}^{-\rho/2}. \tag{24}$$

9.2 Near the origin: $\rho \to 0$

Near $\rho = 0$, the dominant term is

$$-\frac{\ell(\ell+1)}{\rho^2}u.$$

So we approximate

$$\frac{\mathrm{d}^2 u}{\mathrm{d}\rho^2} - \frac{\ell(\ell+1)}{\rho^2}u \approx 0.$$

Try $u \sim \rho^s$. Then

$$s(s-1)\rho^{s-2} - \ell(\ell+1)\rho^{s-2} = 0.$$

Therefore

$$s(s-1) = \ell(\ell+1).$$

This gives

$$s = \ell+1 \quad \text{or} \quad s = -\ell.$$

The $s = -\ell$ solution is singular at the origin. The physical solution is

$$u(\rho) \sim \rho^{\ell+1}. \tag{25}$$

10 The radial ansatz and the Laguerre equation

The two asymptotic results suggest the ansatz

$$u(\rho) = \rho^{\ell+1}\mathrm{e}^{-\rho/2}v(\rho). \tag{26}$$

The factor $\rho^{\ell+1}$ handles the origin. The factor $\mathrm{e}^{-\rho/2}$ handles infinity. The remaining unknown $v(\rho)$ is expected to be a polynomial.

Substituting Eq. (26) into Eq. (22) gives

$$\rho\frac{\mathrm{d}^2 v}{\mathrm{d}\rho^2} + (2\ell+2-\rho)\frac{\mathrm{d}v}{\mathrm{d}\rho} + (\eta-\ell-1)v = 0. \tag{27}$$

This is the associated Laguerre differential equation.

Where quantization enters

The function $v(\rho)$ must not grow so fast that it destroys the exponential decay. This happens only if the power series for v terminates, so v becomes a polynomial. That termination condition quantizes the energy.

10.1 Power series argument

Assume
$$v(\rho) = \sum_{j=0}^{\infty} a_j \rho^j.$$
Then
$$v'(\rho) = \sum_{j=1}^{\infty} j a_j \rho^{j-1}, \qquad v''(\rho) = \sum_{j=2}^{\infty} j(j-1) a_j \rho^{j-2}.$$
Substitute into Eq. (27). Matching powers of ρ gives the recurrence relation
$$a_{j+1} = \frac{j+\ell+1-\eta}{(j+1)(j+2\ell+2)} a_j. \tag{28}$$
For the series to terminate, the numerator must become zero for some non-negative integer $j = N$:
$$N + \ell + 1 - \eta = 0.$$
Thus
$$\eta = N + \ell + 1. \tag{29}$$
We define the principal quantum number
$$\boxed{n = N + \ell + 1.} \tag{30}$$
Therefore,
$$\boxed{\eta = n.} \tag{31}$$
Since $N = 0, 1, 2, \ldots$, we get
$$\boxed{n = 1, 2, 3, \ldots, \qquad \ell = 0, 1, 2, \ldots, n-1.} \tag{32}$$

11 Energy quantization

From Eq. (23),
$$\eta = \frac{\mu\gamma}{\hbar^2 \kappa}.$$
Since $\eta = n$,
$$\kappa = \frac{\mu\gamma}{\hbar^2 n}.$$
But from Eq. (19),
$$\kappa = \frac{\sqrt{-2\mu E}}{\hbar}.$$
Therefore,
$$\frac{\sqrt{-2\mu E}}{\hbar} = \frac{\mu\gamma}{\hbar^2 n}.$$

Square both sides:

$$\frac{-2\mu E}{\hbar^2} = \frac{\mu^2\gamma^2}{\hbar^4 n^2}.$$

Solve for E:

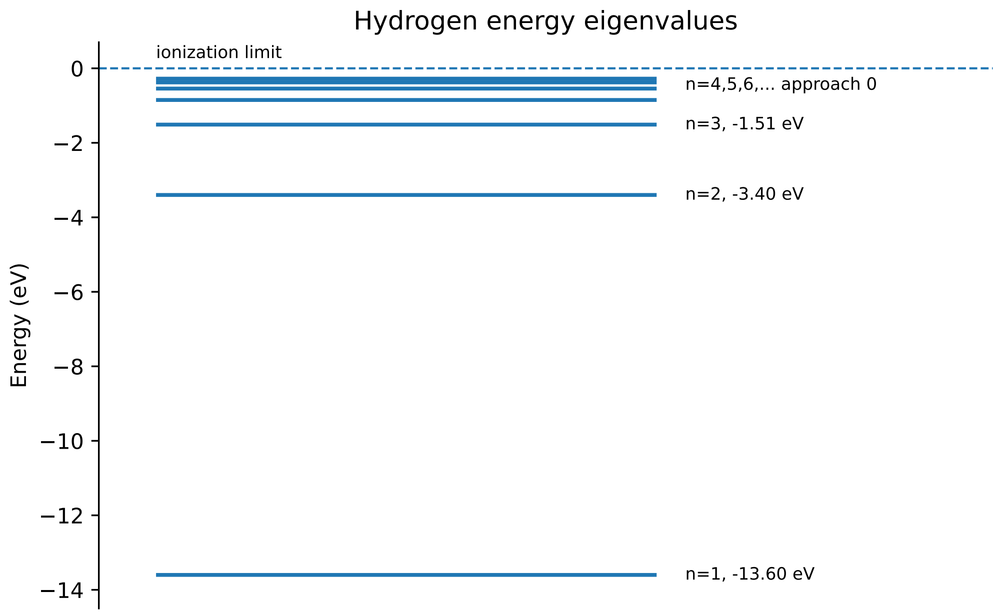
$$\boxed{E_n = -\frac{\mu\gamma^2}{2\hbar^2}\frac{1}{n^2}.} \tag{33}$$

Using $\mu \approx m_e$, this becomes

$$\boxed{E_n = -\frac{13.6\,\text{eV}}{n^2}.} \tag{34}$$

Important conceptual point

Bohr postulated quantization. Schrödinger's equation produces quantization from boundary conditions: finite at the origin, normalizable at infinity, and single-valued in angle.



12 Radial eigenfunctions

The Bohr radius is

$$a_0 = \frac{\hbar^2}{\mu\gamma}. \tag{35}$$

Since $\kappa = 1/(na_0)$, the dimensionless variable is

$$\rho = 2\kappa r = \frac{2r}{na_0}.$$

The normalized radial functions are

$$\boxed{R_{n\ell}(r) = \sqrt{\left(\frac{2}{na_0}\right)^3 \frac{(n-\ell-1)!}{2n[(n+\ell)!]}}\, \mathrm{e}^{-\rho/2}\rho^\ell L^{2\ell+1}_{n-\ell-1}(\rho), \qquad \rho = \frac{2r}{na_0}.} \tag{36}$$

Here L^p_q are associated Laguerre polynomials.

12.1 First examples

$$\begin{aligned}
R_{10}(r) &= 2a_0^{-3/2}\mathrm{e}^{-r/a_0},\\
R_{20}(r) &= \frac{1}{2\sqrt{2}}a_0^{-3/2}\left(2-\frac{r}{a_0}\right)\mathrm{e}^{-r/(2a_0)},\\
R_{21}(r) &= \frac{1}{2\sqrt{6}}a_0^{-3/2}\left(\frac{r}{a_0}\right)\mathrm{e}^{-r/(2a_0)}.
\end{aligned}$$

13 Complete eigenfunctions

The complete normalized eigenfunctions have the form

$$\boxed{\psi_{n\ell m}(r,\theta,\varphi) = R_{n\ell}(r)Y_\ell^m(\theta,\varphi).} \tag{37}$$

The quantum numbers obey

$$\boxed{n = 1,2,3,\ldots, \qquad \ell = 0,1,\ldots,n-1, \qquad m = -\ell,-\ell+1,\ldots,\ell.} \tag{38}$$

Quantum number	Allowed values	Meaning
n	$1,2,3,\ldots$	principal level, energy
ℓ	$0,1,\ldots,n-1$	orbital angular momentum, shape
m	$-\ell,\ldots,\ell$	orientation relative to chosen axis

13.1 Orbital labels

The spectroscopic labels are

$$\ell = 0 \Rightarrow s, \qquad \ell = 1 \Rightarrow p, \qquad \ell = 2 \Rightarrow d, \qquad \ell = 3 \Rightarrow f.$$

Examples:

$$(n,\ell) = (1,0) \Rightarrow 1s, \quad (2,0) \Rightarrow 2s, \quad (2,1) \Rightarrow 2p, \quad (3,2) \Rightarrow 3d.$$

14 Worked derivation of the ground state $1s$

The ground state has

$$n = 1, \qquad \ell = 0, \qquad m = 0.$$

For $\ell = 0$, the radial equation is

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}R}{\mathrm{d}r}\right) + \frac{2\mu}{\hbar^2}\left(E+\frac{\gamma}{r}\right)R = 0. \tag{39}$$

Try

$$R(r) = C\mathrm{e}^{-r/a_0}. \tag{40}$$

Then

$$\frac{\mathrm{d}R}{\mathrm{d}r} = -\frac{1}{a_0}R.$$

So

$$r^2\frac{\mathrm{d}R}{\mathrm{d}r} = -\frac{r^2}{a_0}R.$$

Differentiate:

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}R}{\mathrm{d}r}\right) = -\frac{2r}{a_0}R + \frac{r^2}{a_0^2}R.$$

Divide by r^2:

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}R}{\mathrm{d}r}\right) = -\frac{2}{a_0 r}R + \frac{1}{a_0^2}R.$$

Insert this into Eq. (39):

$$\left[\frac{1}{a_0^2} + \frac{2\mu E}{\hbar^2} + \left(-\frac{2}{a_0} + \frac{2\mu\gamma}{\hbar^2}\right)\frac{1}{r}\right]R = 0.$$

For this to hold for all r, both coefficients must vanish:

$$-\frac{2}{a_0} + \frac{2\mu\gamma}{\hbar^2} = 0 \quad \Rightarrow \quad \boxed{a_0 = \frac{\hbar^2}{\mu\gamma}},$$

and

$$\frac{1}{a_0^2} + \frac{2\mu E}{\hbar^2} = 0 \quad \Rightarrow \quad \boxed{E_1 = -\frac{\hbar^2}{2\mu a_0^2} = -\frac{\mu\gamma^2}{2\hbar^2}.}$$

This is the ground-state energy.

14.1 Normalization of $1s$

For $\ell = 0, m = 0$,

$$Y_0^0(\theta,\varphi) = \frac{1}{\sqrt{4\pi}}.$$

The normalized ground state is

$$\boxed{\psi_{100}(r,\theta,\varphi) = \frac{1}{\sqrt{\pi a_0^3}}\mathrm{e}^{-r/a_0}.} \tag{41}$$

Check normalization:

$$\int |\psi_{100}|^2\,\mathrm{d}V = 1.$$

Since $\mathrm{d}V = r^2\sin\theta\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\varphi$, and ψ_{100} is spherically symmetric,

$$\int |\psi_{100}|^2\,\mathrm{d}V = 4\pi\int_0^\infty \frac{1}{\pi a_0^3}\mathrm{e}^{-2r/a_0}r^2\,\mathrm{d}r.$$

Using

$$\int_0^\infty r^2\mathrm{e}^{-\alpha r}\,\mathrm{d}r = \frac{2}{\alpha^3},$$

with $\alpha = 2/a_0$, we get

$$4\pi\cdot\frac{1}{\pi a_0^3}\cdot\frac{2}{(2/a_0)^3} = 1.$$

15 Radial probability density

The probability density in space is

$$|\psi(r,\theta,\varphi)|^2.$$

But the probability of finding the electron between radii r and $r + \mathrm{d}r$ is

$$P(r)\,\mathrm{d}r = |R_{n\ell}(r)|^2 r^2\,\mathrm{d}r,$$

because the angular part is normalized:

$$\int |Y_\ell^m|^2\,\mathrm{d}\Omega = 1.$$

For the $1s$ state,

$$R_{10} = 2a_0^{-3/2}\mathrm{e}^{-r/a_0}.$$

Thus

$$P_{1s}(r) = r^2|R_{10}|^2 = 4\frac{r^2}{a_0^3}\mathrm{e}^{-2r/a_0}.$$

To find its maximum, differentiate:

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\mathrm{e}^{-2r/a_0}\right) = 2r\mathrm{e}^{-2r/a_0} - \frac{2}{a_0}r^2\mathrm{e}^{-2r/a_0}.$$

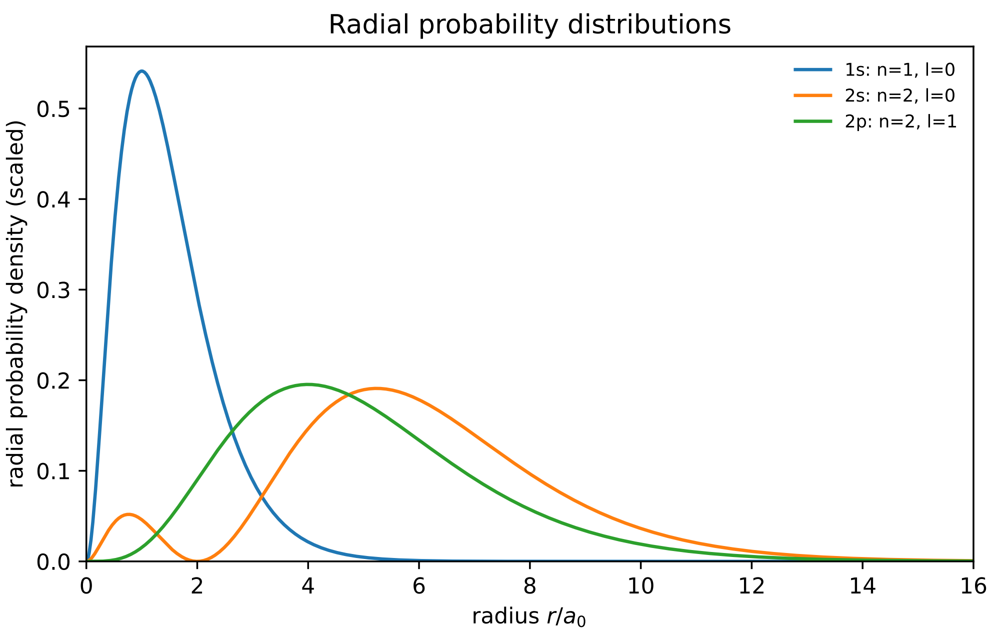
Set equal to zero:

$$2r\left(1 - \frac{r}{a_0}\right)\mathrm{e}^{-2r/a_0} = 0.$$

The nonzero maximum is

$$\boxed{r = a_0.} \tag{42}$$

This explains why the Bohr radius still appears in the Schrödinger picture: it is the most probable radius in the ground state, not a circular orbit.



16 Connection with spectral lines

The energy levels are

$$E_n = -\frac{13.6\,\mathrm{eV}}{n^2}.$$

A photon emitted in a transition $n_i \to n_f$, with $n_i > n_f$, has energy

$$E_\gamma = E_i - E_f.$$

Therefore,

$$E_\gamma = 13.6\,\mathrm{eV}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right).$$

Since

$$E_\gamma = h\nu = \frac{hc}{\lambda},$$

we obtain

$$\boxed{\frac{1}{\lambda} = R_H\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right).} \tag{43}$$

This is the Rydberg formula. The Balmer series is the special case

$$n_f = 2, \qquad n_i = 3, 4, 5, \ldots$$

so

$$\frac{1}{\lambda} = R_H\left(\frac{1}{2^2} - \frac{1}{n_i^2}\right).$$

17 Selection rules, briefly

The energy formula depends only on n, but transitions also depend on angular momentum. For electric dipole radiation, the most important selection rules are

$$\boxed{\Delta\ell = \pm 1, \qquad \Delta m = 0, \pm 1.} \tag{44}$$

Examples:

$$2p \to 1s \quad \text{is dipole-allowed,}$$

while

$$2s \to 1s \quad \text{is not allowed as a simple electric dipole transition.}$$

18 Algorithmic recipe for separation of variables

For students who think algorithmically, the derivation has the following structure.

Separation algorithm

1. Write the stationary equation $\hat{H}\psi = E\psi$.
2. Use spherical coordinates because $V = V(r)$.
3. Try $\psi = R(r)\Theta(\theta)\Phi(\varphi)$.

4. Divide by $R\Theta\Phi$ and multiply by r^2.
5. Separate radial and angular terms with constant $\ell(\ell+1)$.
6. Separate θ and φ terms with constant m^2.
7. Apply angular boundary conditions:

$$m \in \mathbb{Z}, \qquad \ell = 0, 1, 2, \ldots, \qquad |m| \leq \ell.$$

8. Transform the radial equation using $u = rR$.
9. Analyze $r \to 0$ and $r \to \infty$, then use

$$u(\rho) = \rho^{\ell+1} \mathrm{e}^{-\rho/2} v(\rho).$$

10. Require the series for v to terminate.
11. Obtain

$$n = N + \ell + 1, \qquad E_n = -\frac{13.6\,\mathrm{eV}}{n^2}.$$

19 What each quantum number comes from

Quantity	Mathematical origin	Physical meaning
m	Single-valuedness of $\Phi(\varphi) = \mathrm{e}^{\mathrm{i}m\varphi}$	Orientation of angular momentum
ℓ	Finite angular solutions of the associated Legendre equation	Orbital angular momentum / shape
n	Termination of the radial power series	Energy level / shell

20 Common mistakes

Mistake 1: treating r

In the Schrödinger model, $r = a_0$ is the most probable radius for the $1s$ state. It is not a circular trajectory.

Mistake 2: confusing density with radial probability

$|\psi|^2$ is probability density per unit volume. Radial probability includes the spherical shell factor. For an s-state,

$$P(r) = 4\pi r^2 |\psi(r)|^2,$$

or, if angular functions are separated and normalized,

$$P(r) = r^2 |R(r)|^2.$$

Mistake 3: wrong allowed values for fixed n

For a fixed principal quantum number n, the allowed values are

$$\ell = 0, 1, \dots, n-1.$$

For example, if $n = 2$, then $\ell = 0, 1$, so we have $2s$ and $2p$, but no $2d$.

21 Short exercises for the lecture

Exercise 1

Show explicitly that $\Phi(\varphi) = \mathrm{e}^{\mathrm{i}m\varphi}$ is single-valued only if m is an integer.

Exercise 2

For $n = 3$, list all allowed values of ℓ and m. How many states are there if spin is ignored?

Exercise 3

Using

$$E_n = -\frac{13.6\,\mathrm{eV}}{n^2},$$

compute the photon energy for the transition $n_i = 3 \to n_f = 2$.

Exercise 4

For the $1s$ state,

$$P(r) = 4\frac{r^2}{a_0^3}\mathrm{e}^{-2r/a_0}.$$

Differentiate $P(r)$ and show that the maximum occurs at $r = a_0$.

Exercise 5

Explain in one sentence why the energy is quantized in the Schrödinger derivation.

22 Summary

The solution of the hydrogen atom is not just a formula. It is a sequence of constraints:

$$\text{spherical symmetry} \Rightarrow \psi = R\Theta\Phi,$$

$$\text{single-valued angle} \Rightarrow m \in \mathbb{Z},$$

$$\text{finite angular functions} \Rightarrow \ell = 0, 1, 2, \dots, \quad |m| \le \ell,$$

$$\text{normalizable radial function} \Rightarrow n = 1, 2, 3, \ldots,$$

$$\text{energy eigenvalues} \Rightarrow E_n = -\frac{13.6\,\text{eV}}{n^2}.$$

Final message

The Schrödinger equation does not assume Bohr's circular orbits. Instead, it treats the electron as a wave. Quantization appears because only certain wave functions satisfy the mathematical and physical boundary conditions.